Exercise to laboratory 3B: Zonal mean flow and β-plan

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1 Theory (12P)

i. State the necessary assumption that are needed to study Rossby waves. (2P)

ii. Write down the dispersion relation of Rossby waves, the equation for the group velocity and the phase speed. Discuss the dispersion of Rossby waves. (2P)

iii. Discuss in your own words the term dispersion. (1P)

iv. From the equations of questions ii., discuss the role of the wave number k (in relation to the Rossby radius R). Which cases can be distinguished? Describe the cases. (3P)

v. Write down where and how the background flow is introduced in the SWE.(1P)

vi. Describe how the background flow could interfere with the wave propagation and its energy propagation. (2P)

vii. Why do we have to assume reduced gravity in this laboratory? (1P)

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2 Experiments

Set up the model as described in the description for laboratory 3A.

2.1 The role of wave number (10P)

i. List 6 values of kR and the corresponding Lw that sample the characteristic shape of the zonal Rossby wave phase c_x and group c_{gx} speed relations (if $Lw = 1.3\frac{\pi}{k \cdot R}\frac{1}{\sqrt{5}}\frac{\sqrt{g'H}}{f_0}$.) and explain you choice. (1.5P)

ii. Plot the theoretical zonal phase c_x and group c_{gx} speeds as a function of kR and highlight the values that correspond to your choice of i. (1.5P)

iii. Choose three kR values that represents the characteristic regimes of the Rossby wave and provide Hovmoeller diagrams at the center of your y-axis. Estimate the numerical zonal phase and group velocity and plot them as lines into your plots. Add the theoretical zonal phase and group speed as lines into the Hovmoeller diagram. (6P)

iv. Add the numerical the estimates of your three characteristic waves into the plot from ii. (1P) (It is fine if you include one final plot for question ii. and iv. together.)

2.2 The role ot the zonal mean flow (8P)

i. Plot a set of functions for the theoretical c_x and c_g as a function of kR. The background mean flow $U_0 \in [-15, 15]$ should be your parameter that creates the set of function. From that fix your kR you want to investigate further in your experiments for a variation of U_0 and plot the theoretical u_g and u_x as a function of U_0. Define at least three values of U_0 you want to investigate and motivate your choice. (4.5P)

ii. Plot Hovmoeller diagrams of at least three representative waves on the beta-plane with background flow. Add the theoretical zonal phase and group speed to the plot. (2P)

iii. Plot your waves shown in i. in xy-cross sections at a appropriate time. (1.5P)

Good luck!